UNIVERSITY OF RWANDA

CST- COMPUTER ENGINEERING

LEVEL IV

GAKO CAMPUS

MOBILE COMMUNICATION

Mobile Communications – Unit 2 Mini Projects Report

ASSIGNIMENT On 27 OCT 2026

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Github link: **https://github.com/filsniyonzima/mc_unit2_mp.git**

Mini Project 2.1: Free-Space Path Loss and Digital Modulation Performance

1. Objective and Method

This project links two central Unit 2 results, the Friis free-space path-loss equation and the Shannon-Hartley capacity theorem, to simulated behaviour. Everything was implemented in Python 3 (NumPy, SciPy, Matplotlib) in a single script, mp2_1.py, with a fixed random seed (np.random.seed(0)) so results are reproducible. Part (a) evaluates received power versus distance; Part (b) measures the bit error rate (BER) of BPSK and 16-QAM over an AWGN channel by Monte Carlo simulation, and additionally of QPSK and 64-QAM (lab tasks 1 and 3).

2. Part (a): Friis Path Loss

Received power was computed as $P_RX = P_TX + G_TX + G_RX - 20\cdot log10(d_km) - 20\cdot log10(f_MHz) - 32.44$ with P_TX = 43 dBm, G_TX = 15 dBi, G_RX = 0 dBi, d swept over 0.05–10 km (500 points) and f ∈{900, 1900, 3500} MHz. Figure 1 shows the result: each curve is a straight line against log-distance, and the higher the carrier frequency, the lower the curve at every distance.

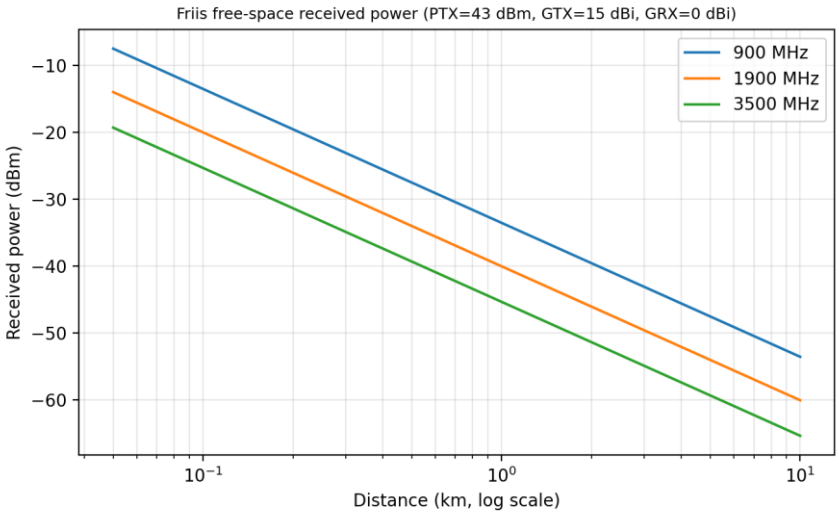


Figure 1: Received power versus distance for three carrier frequencies.

The 6 dB-per-doubling rule was verified numerically (printed by the script). Because distance and frequency both sit inside a 20·log10(·) term, doubling either costs 20·log10(2) = 6.02 dB:

Check	P_RX before (dBm)	P_RX after (dBm)	Drop (dB)
Distance 1 km → 2 km (900 MHz)	−33.52	−39.55	6.021
Frequency 900 → 1800 MHz (1 km)	−33.52	−39.55	6.021

The measured drop equals the theoretical 6.021 dB exactly, since Friis is a closed-form law. The gaps between curves follow the same logic: 1900 MHz sits 20·log10(1900/900) = 6.49 dB below 900 MHz (−33.99 vs −27.50 dBm at 0.5 km), and 3500 MHz sits 5.30 dB below 1900 MHz. At 5 km the received powers are −47.5, −54.0 and −59.3 dBm respectively. The practical consequence is that, for the same antennas and power, a 3.5 GHz cell has noticeably less range than a 900 MHz cell, which is why low bands are preferred for coverage.

3. Part (b): BER of BPSK, QPSK, 16-QAM and 64-QAM over AWGN

A random bitstream of N = 1.8 × 10^6 bits was mapped to unit-average-energy symbols: BPSK (0 → −1, 1 → +1) and Gray-coded square QPSK, 16-QAM (levels −3, −1, +1, +3) and 64-QAM. Complex Gaussian noise was added with N0 = Es / (k·Eb/N0), where k is bits per symbol, so that every scheme is compared at the same energy per bit. Detection used the minimum-distance rule. Eb/N0 was swept from 0 to 16 dB in 1 dB steps and BER was taken as errors divided by total bits. The mappers were first checked with noiseless round-trip tests and a unit-energy assertion.

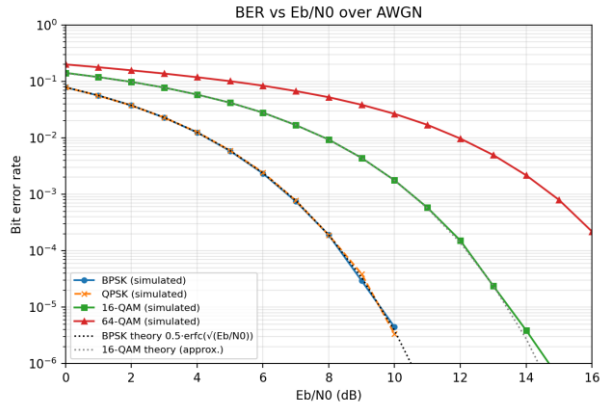


Figure 2: Simulated BER vs Eb/N0 with theory overlays.

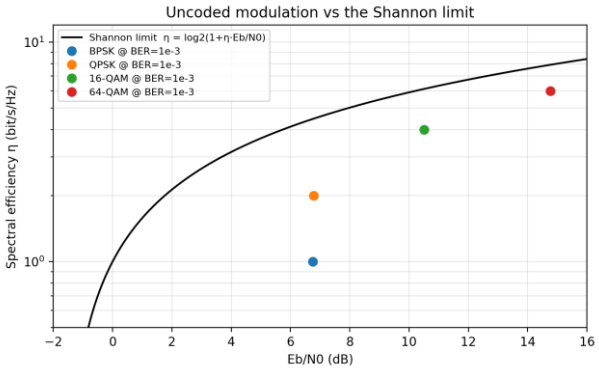


Figure 3: Operating points at BER = 10^{-3} vs the Shannon limit.

Validation. The simulated BPSK curve (Figure 2) follows the theoretical 0.5·erfc(√(Eb/N0)) curve to within 0.06 decades wherever BER > 10^{-5}, and the simulated 16-QAM curve follows the standard (3/8)·erfc(√(0.4·Eb/N0)) approximation. QPSK lands on top of BPSK, as expected: it is two independent BPSK streams on the I and Q axes, so it carries 2 bits/symbol at the same per-bit Eb/N0. All curves show the typical waterfall shape, shifted right as the constellation grows (points with zero observed errors are omitted from the log plot).

4. SNR Gap and Link to Shannon-Hartley

Reading the curves at a target BER of 10^{-3} (log-domain interpolation) gives:

Scheme	Eb/N0 (dB)	Es/N0 (dB)	Shannon min Eb/N0 (dB)	Gap to Shannon (dB)
BPSK (1 b/sym)	6.75	6.75	0.00	6.75
QPSK (2 b/sym)	6.78	9.79	1.76	5.02
16-QAM (4 b/sym)	10.51	16.53	5.74	4.77
64-QAM (6 b/sym)	14.77	22.55	10.21	4.56

Numeric gap. 16-QAM needs 3.76 dB more Eb/N0 than BPSK for BER = 10^{-3}, which is 9.79 dB more SNR per symbol (Es/N0). The lab manual's "roughly 8–10 dB" figure matches the per-symbol SNR gap; the per-bit gap is smaller because 16-QAM spreads each symbol's energy over four bits, a distinction worth stating explicitly in a submission.

Explanation. Shannon-Hartley, C = B·log2(1 + SNR), implies that a spectral efficiency η requires at least SNR = 2^η − 1, i.e. Eb/N0 ≥ (2^η − 1)/η. Moving from 1 to 4 bits per symbol therefore raises the minimum required SNR from 0 dB to 11.76 dB: each extra bit per channel use costs about 3 dB at high SNR, because capacity grows only logarithmically with SNR. The simulation shows the physical mechanism: a denser constellation packs more points into the same energy, so the minimum distance between neighbours shrinks and the same noise causes more errors. The measured Eb/N0 increases (6.78 → 10.51 → 14.77 dB for 2, 4, 6 bits/symbol) track the Shannon increases (1.76 → 5.74 → 10.21 dB) closely, +3.7 and +4.3 dB against +4.0 and +4.5 dB. The gap is consistent with Shannon-Hartley but not numerically predicted by it, since the theorem describes an idealized capacity-achieving code with arbitrarily long blocks, whereas uncoded modulation always operates roughly 4.5–5 dB above the ceiling at BER = 10^{-3}. Closing that gap is the job of channel coding. (BPSK appears further from the limit in Figure 3 only because it is plotted at η = 1 under the complex-baseband convention, although it uses a single signal dimension; counted per real dimension it matches QPSK.)

Throughput proxy (lab task 6). Using k·(1 − BER) bits/symbol, 16-QAM delivers 3.77 at 4 dB but 4.00 at 12 dB, while BPSK is already at 1.00 by 8 dB: higher-order schemes only pay off once SNR is high enough, the same trend as the Shannon capacity curve.

Mini Project 2.2: OFDM Waveform Generator with a Cyclic Prefix Visualizer

1. Objective and System Design

The goal was to build the smallest complete OFDM link that makes the effect of the cyclic prefix (CP) directly observable. The script mp2_2.py (Python 3, NumPy, Matplotlib) implements the chain: random bits → Gray-coded 16-QAM → N = 64 subcarriers → IFFT → CP insertion → multipath FIR channel → (optional AWGN) → CP removal → FFT → one-tap zero-forcing equaliser → hard detection. A stream of 400 consecutive OFDM symbols is convolved with the channel as one serial signal, so each symbol's tail genuinely leaks into the next one. The first symbol is excluded from the metrics because it has no predecessor. Damage is measured with symbol error rate (SER) and the mean-squared error (MSE) between recovered and transmitted constellation points (unit average symbol energy).

2. Channel Model and Verification

The synthetic channel is a 3-tap FIR filter, normalised to unit power. Its maximum excess delay is 7 samples, so a CP of at least 7 samples should be sufficient.

Tap	Delay (samples)	Gain magnitude	Gain phase
1	0	0.855	0°
2	3	0.453	+52°
3	7	0.252	−92°

Each block was unit-tested before the full system was assembled, as the project brief requires:

Self-check	Result
16-QAM mapper noiseless round trip; mean symbol energy	Pass; energy = 0.996
FFT of own noiseless IFFT output vs original subcarriers	Max error 6.3×10^{-16} (pass)
CP (16 samples) equals the tail of the OFDM symbol	Pass
Channel frequency response |H| over 64 subcarriers	min 0.171, max 1.553 (no deep null)

3. Results: Sufficient versus Insufficient CP

On the identical channel realisation, the three receivers give very different results (noiseless, 16-QAM). With a 16-sample CP the equalised constellation collapses back onto the 16 ideal points exactly; with a 4-sample CP the points are visibly smeared; with no CP the clusters merge and the SER exceeds 10% (Figure 1).

Receiver	CP length	SER	MSE
Sufficient CP	16	0.0000	1.4×10^{-31}
Too-short CP	4	0.0119	1.5×10^{-2}
No CP	0	0.1017	9.5×10^{-2}

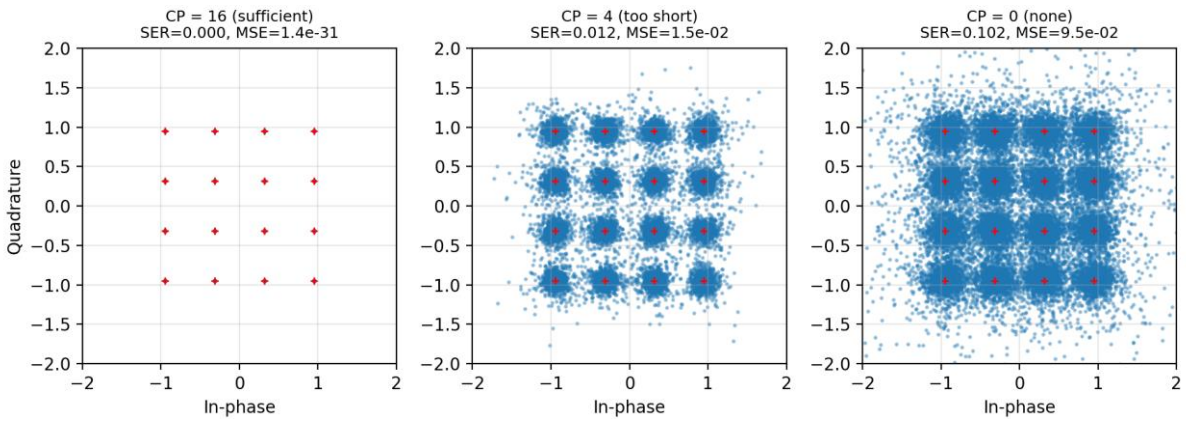


Figure 1: Equalised constellations (red + = ideal points): sufficient CP, too-short CP, and no CP.

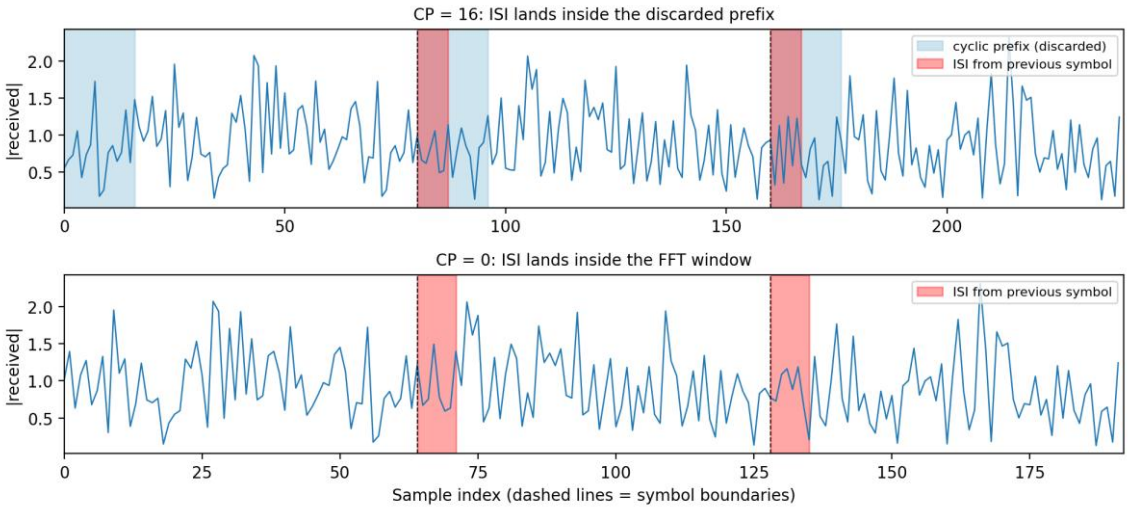


Figure 2: Received waveform |y[n]| for three consecutive symbols. Red: samples contaminated by the previous symbol; blue: the CP that is discarded.

4. Error Rate versus CP Length

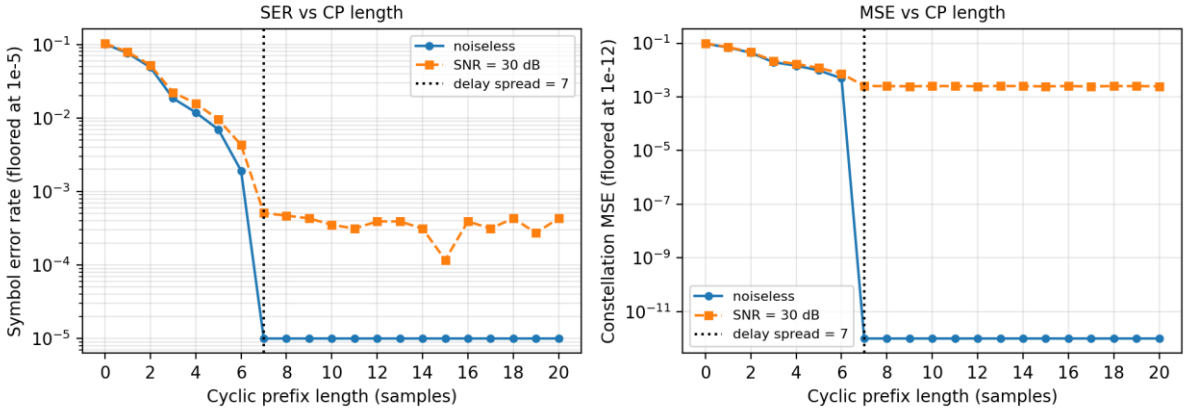


Figure 3: SER and MSE as the CP is swept from 0 to 20 samples (channel delay spread = 7 samples).

In the noiseless sweep the error falls steadily as the CP grows (SER 0.102 at CP = 0, 0.012 at 4, 0.002 at 6) and reaches exactly zero at CP = 7, the channel's maximum delay; MSE drops from 10^{-2} to numerical zero (10^{-31}). Beyond 7 nothing improves. With 30 dB of AWGN added, the curves flatten at a constant noise floor (SER ≈ 4×10^{-4}, MSE ≈ 2.5×10^{-3}) that is set by noise and zero-forcing noise enhancement, not by ISI. The approach to the threshold is gradual rather than a cliff because the last tap is weak (0.25), so when the CP is only slightly too short just a few samples are contaminated and by a small amount.

5. Explanation: The CP's Dual Role

Guard interval. A delayed path copies the end of the previous symbol into the start of the current one, but only for as many samples as the delay spread (7 here, the red regions in Figure 2). When the CP is at least that long, all contaminated samples fall inside the prefix, which the receiver discards, so no ISI reaches the FFT window. With no CP (lower panel) the same red samples sit inside the window and corrupt the data.

Cyclic extension. Because the prefix is a copy of the symbol's tail, linear convolution with the channel looks like circular convolution inside the FFT window. Hence $Y_k = H_k \cdot X_k$ on every subcarrier, which is why a single complex division per subcarrier recovers the data and orthogonality between subcarriers is preserved. A CP that is too short breaks this circular structure, causing inter-carrier interference on top of ISI, which is the smearing seen in Figure 1.

Overhead trade-off. Once CP ≥ 7 the error rate plateaus, so extra prefix only wastes capacity: efficiency N/(N + CP) is 0.90 at CP = 7 but only 0.80 at CP = 16. A CP slightly longer than the expected delay spread is the best design point.

Mini Project 2.3: Comparing WPAN, WLAN and WMAN Technologies on Range, Throughput and Power

1. Objective and Method

This project turns the qualitative Unit 2 comparison of wireless networks by geographic scope into a small quantitative tool. The script mp2_3.py (Python 3, pandas, NumPy, Matplotlib) holds a sourced dataset of five technologies across the three network classes, computes derived metrics (including free-space path loss from the Friis equation of Project 2.1), draws the trade-off charts, and checks each technology against three deployment scenarios with explicit numbered requirements. Every specification figure carries a source number [n] listed in Section 6.

2. Dataset

Technology	Class	Band (GHz)	Range: typical → max	Rate (Mbps): typical / peak	Tx power (dBm): infra / client	Access / PHY	Sources
Bluetooth 5 (BLE)	WPAN	2.4	40 m → 200 m	1.4 / 2	0 / 0	FHSS, GFSK	[1][2][3]
Wi-Fi 5 (802.11ac)	WLAN	5.5	27 m → 300 m	867 / 1,730	18 / 18	OFDM, CSMA/CA	[4][8][10]
Wi-Fi 6 (802.11ax)	WLAN	5.5	30 m → 120 m	1,201 / 2,402	21 / 23	OFDMA, CSMA/CA	[4][5][6][7]
WiMAX 802.16e	WMAN	2.5	3 km → 10 km	5.3 / 70	43 / 23	Scalable OFDMA	[9][11][12]
WiMAX 802.16m	WMAN	2.5	3 km* → 10 km	100 / 1,000	43* / 23*	Scalable OFDMA	[9][11][13]

Assumptions and caveats. Range is strongly environment-dependent: “typical” is an indoor (WPAN/WLAN) or urban-mobile (WMAN) figure and “max” is outdoor line-of-sight or fixed-link. Wi-Fi 6 indoor range disagrees between sources (30 m [7] vs about 15 m at 5 GHz [8]); I used 30 m. Peak rates are theoretical PHY rates for the configuration shown (e.g. 2×2 MIMO, 160 MHz) and are shared among users; “typical” is a common configuration, except WiMAX 16e (5.3 Mbps, field-measured [12]) and BLE (application throughput [3]). For 802.16m, the lower range bound and Tx power (*) were assumed equal to 802.16e because no separate figures were found. Tx power is the access-point/base-station value (conducted or EIRP depending on source); 5.5 GHz is the 5 GHz band mid-point used for path loss. Much of the WiMAX literature dates from 2006–2010.

3. Derived Metrics

Technology	Tx power (mW)	FSPL at typical range (dB)	FSPL at max range (dB)	Peak rate ÷ range (Mbps/m)	Peak rate × range (Mbps·km)
Bluetooth 5	1	72.1	86.1	0.05	0.08
Wi-Fi 5	63	75.9	96.8	64.1	46.7
Wi-Fi 6	126	76.8	88.8	80.1	72.1
WiMAX 802.16e	19,953	109.9	120.4	0.02	210
WiMAX 802.16m	19,953	109.9	120.4	0.33	3,000

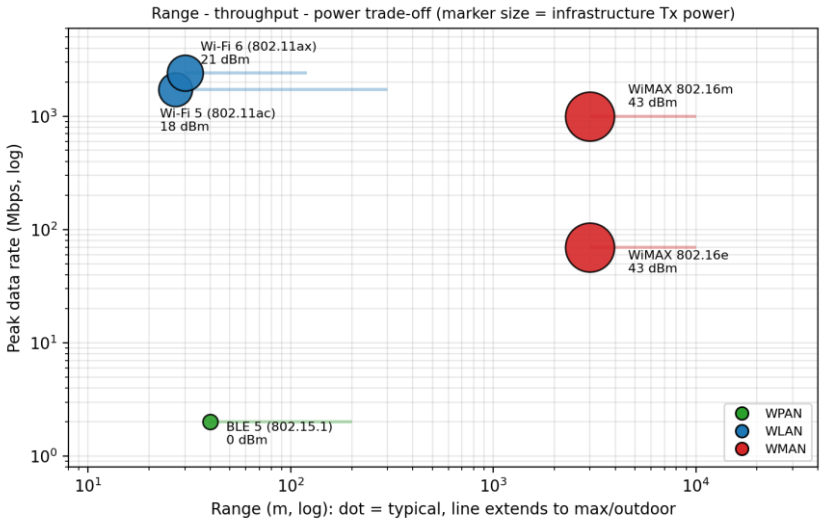


Figure 1: Range versus peak data rate on log axes; marker size scales with infrastructure Tx power.

Three patterns stand out. First, range is bought largely with power: Tx power spans 43 dB (0 to 43 dBm, about 20,000×) while free-space path loss at typical range spans 38 dB (72 to 110 dB), so WiMAX's base-station power almost exactly pays for its extra distance, helped by its lower carrier (20·log10(5500/2500) = 6.8 dB less loss than

5.5 GHz at equal distance). Second, Wi-Fi 6 has the best rate per metre of range (80 Mbps/m) because it concentrates very high rates over short links, whereas WiMAX 802.16m has the highest rate × range product (3,000 Mbps·km). Third, Bluetooth is lowest on both measures but uses 18 to 43 dB less transmit power than the others, which is what its battery-powered use cases require. No technology wins on all three axes, which is the trade-off Figure 1 shows.

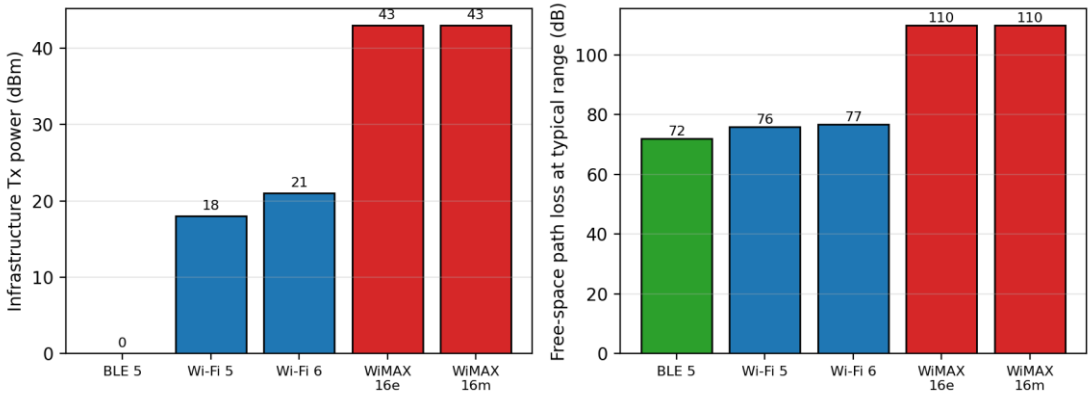


Figure 2: Infrastructure Tx power and free-space path loss at each technology's typical range.

4. Deployment Scenarios

The requirements below are design values chosen for this exercise (they are assumptions, not measurements). Scenarios were checked against the dataset on range, rate and power; for S1 and S2 the typical rate is used, and for S3 (fixed subscribers) the stationary peak rate, with S3 range judged against the fixed-link maximum.

Scenario	R1 Coverage	R2 Devices / load	R3 Throughput	R4 Power / cost constraint
S1 Smart home	15 m radius, indoors	50 sensors	10 kbps each = 0.5 Mbps aggregate	Coin cell, 2+ years; Tx ≤ 10 dBm
S2 Campus	0.2 km² site, 25 m cells	1,500 users, about 30 per AP	10 Mbps each = 300 Mbps per cell, 15 Gbps site	Mains-powered APs; unlicensed spectrum preferred
S3 Metro fixed wireless	6 km radius	300 fixed homes	5 Mbps each, 10:1 contention = 150 Mbps per site	Mains-powered base station; spectrum licence acceptable

Technology	S1 Smart home	S2 Campus	S3 Metro FWA
Bluetooth 5	Pass	Fail: rate	Fail: range, rate
Wi-Fi 5 (802.11ac)	Fail: power	Pass	Fail: range
Wi-Fi 6 (802.11ax)	Fail: power	Pass	Fail: range
WiMAX 802.16e	Fail: power	Fail: rate	Fail: rate (3 sectors needed)
WiMAX 802.16m	Fail: power	Fail: rate	Pass

Table: Requirement check from the script (Pass = meets range, rate and power requirements).

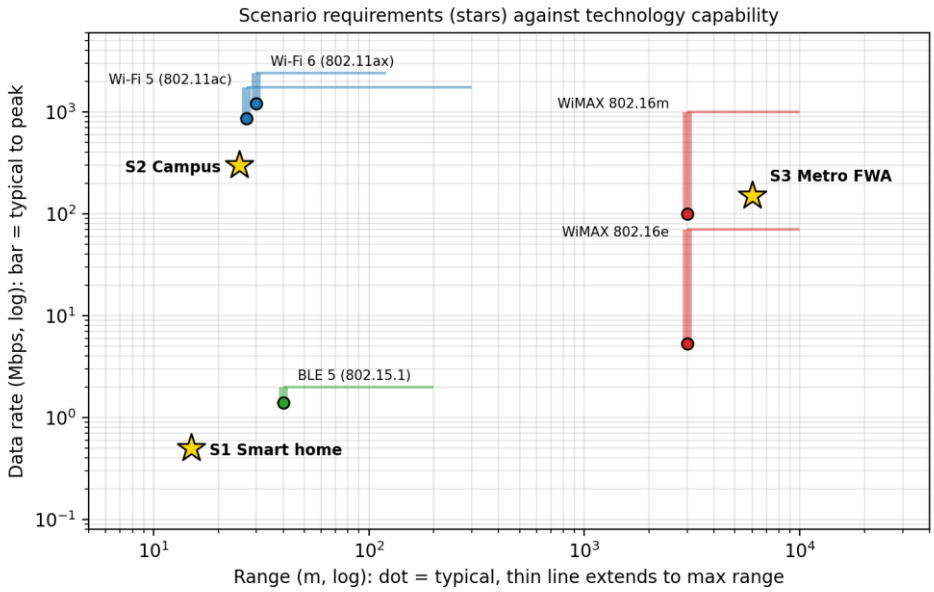


Figure 3: Scenario requirements (stars) against each technology's typical-to-peak rate bar and range.

5. Recommendations

S1, Smart home: Bluetooth 5 (BLE). The needs are modest: 15 m range, 0.5 Mbps aggregate, minimal power. BLE gives 40 m indoor range (2.7× margin) and 1.4 Mbps throughput (2.8× margin) at 0 dBm (1 mW), whereas a Wi-Fi client transmits at 18 to 23 dBm, 63 to 200 times more power. Wi-Fi meets the range and rate numbers but exceeds the power budget, and WiMAX's 3 km range is irrelevant here. Secondary factors: BLE's frequency hopping over 40 channels [1] helps coexistence in the crowded 2.4 GHz band shared with Wi-Fi; a gateway (typically Wi-Fi) is still needed for internet access, so the realistic design is a hybrid; and BLE chips are cheap, license-free and already in phones.

S2, Campus: Wi-Fi 6 (802.11ax). Each 25 m cell must carry 300 Mbps; Wi-Fi 6 offers 1,201 Mbps typical (4× margin) against Wi-Fi 5's 867 Mbps (2.9×), while WiMAX 16e (5.3 typical, 70 peak) and BLE cannot come close. The site needs about 102 cells for coverage (200,000 m² ÷ 1,963 m², ideal, no overlap) but only about 12.5 APs for capacity (15,000 Mbps ÷ 1,201), so coverage, not capacity, sets the AP count, and Wi-Fi 6 and Wi-Fi 5 both pass. Wi-Fi 6 is preferred for its OFDMA multi-user scheduling [5], higher peak rate and similar Tx power (21 versus 18 dBm). Secondary factors: unlicensed spectrum means no licence cost, mass-market client support, but roughly 100 APs need cabling/PoE backhaul and roaming planning, which usually dominate cost; WiMAX would need about 214 base stations for capacity alone (15,000 ÷ 70).

S3, Metro fixed wireless: WiMAX 802.16m (802.16e with sectors on paper). Only the WMAN class reaches 6 km: 802.16 quotes up to 10 km for fixed links [11] (1.7× margin), against 120 to 300 m outdoors for Wi-Fi. The base station needs 150 Mbps; 802.16m offers 1 Gbps stationary peak (6.7×), whereas 802.16e's 70 Mbps peak would need at least 3 sectors, and its field-measured 5.3 Mbps [12] is a reminder that peak is not delivered throughput. Path loss at 6 km and 2.5 GHz is about 116 dB, supportable with +43 dBm base-station power [9]. Wi-Fi is not a total non-starter: directional 5 GHz bridges reach 5 km at 867 Mbps [15], but only as point-to-point links, not as a 300-home coverage network. Secondary factors: WiMAX typically needs licensed spectrum [10], and deployment maturity is a serious complication: major operators retired their WiMAX networks in favour of LTE (Sprint closed its network in 2015 [14]), so equipment and support are scarce. In practice a new metro link would use LTE/5G fixed wireless or licensed microwave, but among the Unit 2 technologies WiMAX is the one that fits the numbers.

6. Sources

[1] Electronic Design, '5 Things You Must Know About the New Bluetooth 5' - electronicdesign.com/technologies/communications/wireless/article/21802861/5-things-you-must-know-about-the-new-bluetooth-5

[2] BZZ course wiki, LU07b Bluetooth (BT5: ~40 m indoors / ~200 m outdoors at 1 Mbit/s) - wiki.bzz.ch/en/modul/m286/learningunits/lu07/bluetooth

[3] Android Authority, 'Bluetooth 5: speed, range and more' (BLE 5 2 Mbps PHY, ~1.4 Mbps throughput) - androidauthority.com/bluetooth-5-speed-range-762369/

[4] Intel Support, 'Different Wi-Fi Protocols and Data Rates' (802.11ax 2.4 Gbps, 802.11ac wave 2 1.73 Gbps, 866 Mbps 2x2 80 MHz) - intel.com/content/www/us/en/support/articles/000005725.html

[5] Aruba 500H series datasheet (HE80 2SS 1.2 Gbps; +21 dBm 5 GHz) - open-vision.ru/catalog/network-equipment/besprovodnyie-seti/aruba-500h-series/

[6] Intel Wi-Fi 6E AX210 technical specifications (5 GHz client 23 dBm EIRP max) - onlinehelp.ncrvoyix.com/Retail/SafetyCompliance/B005-0000-1589/Topics/RFIStatements/AX210/AX210TechSpec.htm

[7] Electronics Notes, Wi-Fi standards infographic (Wi-Fi 6: ~30 m indoor, ~120 m outdoor) - electronics-notes.com/articles/summary-infographics/wifi-80211/wifi-standards-infographic-v3.pdf

[8] L-com, '802.11 Wireless Standards' (11ac ~90 ft indoor, ~1000 ft outdoor; 11ax 50 ft / 900 ft at 5 GHz) - l-com.com/images/downloadables/Technical-Briefs-And-Articles/802-11-Wireless-Standards.pdf

[9] EDN, WiMAX power profile (BS ~+43 dBm / 20 W, MS ~+23 dBm) - edn.com/?p=4005852

[10] EE Times India, 'The File', 1 Dec 2008 (Wi-Fi AP/client typically +18 dBm; WiMAX BS +43 dBm) - nxtbook.com/nxtbooks/emedia/eetindia_20081201/index.php?startid=6

[11] RF Wireless World, 'Understanding IEEE 802.16 WiMAX Standards' (16e: up to 70 Mbps fixed, 3-5 km mobile, up to 10 km fixed; 16m: 1 Gbps stationary / 100 Mbps mobile, up to 10 km) - rfwireless-world.com/articles/understanding-ieee-802-16-wimax-standards

[12] arXiv:1605.00889, 'The Role of communication and network technologies in vehicular applications' (802.16e up to 70 Mbps, ~10 km; measured mean 2 and 5.3 Mbps) - arxiv.org/pdf/1605.00889

[13] itwissen.info, 'IEEE 802.16m' (1 Gbit/s nomadic, 100 Mbit/s at high mobility) - itwissen.info/en/IEEE-802-DOT-16m-119759.html

[14] FierceWireless, 'Sprint to shutter WiMAX network around Nov. 6, 2015' - fierce-network.com/wireless/sprint-to-shutter-wimax-network-around-nov-6-2015

[15] TP-Link Omada 5 GHz 867 Mbps wireless bridge (up to 5 km point-to-point) - omadanetworks.com/ca/business-networking/omada-wifi-wireless-bridge/